

Homework6

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1 Question1

(a)

$\int \varphi(u) du = \frac{3}{2} \Rightarrow$ normalized kernel $\frac{2}{3}\varphi(u)$.

$$\hat{p}(x | \omega_1) = \frac{1}{3} \sum_{x_j \in \{0,2,5\}} \frac{2}{3} \varphi(x-x_j) = \frac{2}{9} \sum_{x_j \in \{0,2,5\}} \varphi(x-x_j), \quad \hat{p}(x | \omega_2) = \frac{1}{3} \sum_{x_j \in \{4,7\}} \varphi(x-x_j).$$

Trapezoids on $(x_j - 1, x_j + 1)$, plateau $[x_j - \frac{1}{2}, x_j + \frac{1}{2}]$; peak heights $\frac{2}{9}$ (ω_1) and $\frac{1}{3}$ (ω_2).

(b)

$$\hat{P}(\omega_1) = \frac{3}{5}, \quad \hat{P}(\omega_2) = \frac{2}{5}.$$

(c)

$$\hat{P}(\omega_1) \cdot \frac{2}{9} = \hat{P}(\omega_2) \cdot \frac{1}{3} = \frac{2}{15} \Rightarrow \text{compare } \sum \varphi \text{ directly.}$$

Overlap only on $(4, 5)$: $\varphi(x-5) = \varphi(x-4) \Rightarrow x^* = 4.5$.

$$\mathcal{R}_1 = (-1, 3) \cup (4.5, 6), \quad \mathcal{R}_2 = (3, 4.5) \cup (6, 8), \quad \text{boundaries } x = 3, 4.5, 6$$

2 Question2

Transformed points:

$$\mathbf{u} = [\varphi_1, \varphi_2]^T$$

$$\begin{aligned} \mathbf{x}_1 &= (1, 0) \rightarrow (1, -2), & \mathbf{x}_2 &= (0, 1) \rightarrow (4, -5), & \mathbf{x}_3 &= (0, -1) \rightarrow (4, -1) & (z = -1) \\ \mathbf{x}_4 &= (-1, 0) \rightarrow (5, -2), & \mathbf{x}_5 &= (0, 2) \rightarrow (7, -7), & \mathbf{x}_6, \mathbf{x}_7 &\rightarrow (7, 1) & (z = +1) \end{aligned}$$

Optimal hyperplane:

Class -1 has $u_1 \leq 4$, class $+1$ has $u_1 \geq 5$. Support vectors: $(4, -5), (4, -1)$

and $(5, -2)$; the projection of $(5, -2)$ onto the segment $u_1 = 4$ is interior, so the maximum-margin hyperplane is vertical, midway:

$$u_1 = 4.5 \quad \Longleftrightarrow \quad g(\mathbf{u}) = 2u_1 - 9 = 0, \quad \mathbf{w} = [2, 0]^T, \quad b = -9,$$

canonical: $g = \mp 1$ at the support vectors;

$$\text{margin} = \frac{2}{\|\mathbf{w}\|} = 1$$

Decision rule: $z = \text{sign}(u_1 - 4.5)$. In the original space: $x_2^2 - 2x_1 - 1.5 = 0$.

3 Question3

(a)

Dual problem

Kernel matrix $K_{ij} = (\mathbf{x}_i^T \mathbf{x}_j + 1)^2$ and labels $\mathbf{z} = (-1, 1, 1, 1)^T$:

$$K = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 4 & 1 & 0 \\ 1 & 1 & 4 & 1 \\ 1 & 0 & 1 & 4 \end{pmatrix}, \quad \max_{\lambda} \sum_{i=1}^4 \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j z_i z_j K_{ij} \quad \text{s.t.} \quad \sum_i z_i \lambda_i = 0, \quad \lambda_i \geq 0.$$

(b)

Solution:

Symmetry ($\mathbf{x}_2 \leftrightarrow \mathbf{x}_4$) gives $\lambda_2 = \lambda_4 = a$, $\lambda_3 = b$; the constraint gives $\lambda_1 = 2a + b$.

Substituting,

$$Q(a, b) = 4a + 2b - \frac{1}{2}(4a^2 + 3b^2) \Rightarrow \frac{\partial Q}{\partial a} = 4 - 4a = 0, \quad \frac{\partial Q}{\partial b} = 2 - 3b = 0,$$

$$\lambda_1 = \frac{8}{3}, \quad \lambda_2 = \lambda_4 = 1, \quad \lambda_3 = \frac{2}{3} \quad (\text{all } \geq 0, \text{ all points are support vectors}).$$

(c)

Since $\mathbf{w} = \sum_i \lambda_i z_i \boldsymbol{\varphi}(\mathbf{x}_i)$,

$$g(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\varphi}(\mathbf{x}) + w_0 = \sum_{i=1}^N \lambda_i z_i \boldsymbol{\varphi}(\mathbf{x}_i)^T \boldsymbol{\varphi}(\mathbf{x}) + w_0 = \sum_{i=1}^N \lambda_i z_i \kappa(\mathbf{x}_i, \mathbf{x}) + w_0.$$

(d)

For $\lambda_j \neq 0$, KKT gives $z_j(\mathbf{w}^T \mathbf{u}_j + w_0) = 1$, so $w_0 = 1/z_j - \sum_i \lambda_i z_i \kappa(\mathbf{x}_i, \mathbf{x}_j)$.

Take $\mathbf{x}_j = \mathbf{x}_2$:

$$\sum_i \lambda_i z_i \kappa(\mathbf{x}_i, \mathbf{x}_2) = -\frac{8}{3}(1) + 1(4) + \frac{2}{3}(1) + 1(0) = 2$$

$$w_0 = 1 - 2 = -1,$$

(e)

Decision boundary:

With $\mathbf{x} = (x_1, x_2)^T$:

$$g(\mathbf{x}) = -\frac{8}{3} + (x_1 + 1)^2 + \frac{2}{3}(1 - x_2)^2 + (1 - x_1)^2 - 1 = 2x_1^2 + \frac{2}{3}x_2^2 - \frac{4}{3}x_2 - 1.$$

$$g(\mathbf{x}) = 0 \iff 3x_1^2 + (x_2 - 1)^2 = \frac{5}{2}$$

an ellipse centered at $(0, 1)$ with semi-axes $\sqrt{5/6}$ (x_1) and $\sqrt{5/2}$ (x_2): $\mathbf{x}_1$ inside ($z = -1$), $\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$ outside ($z = +1$). Check: $g(\mathbf{x}_1) = -1$, $g(\mathbf{x}_2) = g(\mathbf{x}_3) = g(\mathbf{x}_4) = +1$.

4 Question4

(a) Plot 3, (b) Plot 4, (c) Plot 1, (d) Plot 6, (e) Plot 5;
Plot 2 has no match.

Explanation:

(a) 3: tiny C tolerates violations $\Rightarrow$ wide margin, several misclassified points.

(b) 4: large C penalizes violations $\Rightarrow$ clean linear separation, tight margin.

(c) 1: degree-2 polynomial kernel $\Rightarrow$ quadratic (conic) boundary; the ellipse is the only conic.

(d) 6: RBF with large $\gamma = 5 \Rightarrow$ narrow Gaussians, overfit contour hugging one class.

(e) 5: RBF with small $\gamma = \frac{1}{5} \Rightarrow$ broad Gaussians, smooth gently curved boundary.

Plot 2: unmatched — too wiggly for linear or broad RBF, not a conic, not a tight closed contour.

5 Question5

(a)

Download the data.

Take 70 percent of the data randomly as the training set.

(bi)

Do setup first

Research exact match and hamming score/ loss methods for evaluating multi-label classification.

Using them in evaluating the classifiers in this problem.

(bii)

Normalized attributes:

Label	Best (C, γ)	10-fold CV acc.
Family	(100, 0.01)	0.9899
Genus	(100, 0.01)	0.9865
Species	(100, 0.01)	0.9869
Test metric		Value
Exact match		0.9866
Hamming score		0.9910
Hamming loss		0.0090
Family / Genus / Species acc.		0.9931 / 0.9898 / 0.9903

Raw attributes:

Label	Best (C, γ)	10-fold CV acc.
Family	(10, 1.0)	0.9915
Genus	(10, 1.0)	0.9885
Species	(10, 1.0)	0.9873
Test metric		Value
Exact match		0.9875
Hamming score		0.9921
Hamming loss		0.0079
Family / Genus / Species acc.		0.9926 / 0.9921 / 0.9917

Both reach essentially the same test performance (0.9875 vs. 0.9866).
The optimal kernel width shifts from $\gamma = 1$ to $\gamma = 0.01$ with a larger C (two equivalent operating points rather than different models).
Normalization does not help because the MFCC coefficients already share a comparable scale.

(biii)

Label	Best C	10-fold CV acc.
Family	1	0.9343
Genus	100	0.9533
Species	100	0.9573
Test metric		Value
Exact match		0.9171
Hamming score		0.9494
Hamming loss		0.0506
Family / Genus / Species acc.		0.9393 / 0.9491 / 0.9597

Gaussian kernel vs. $\mathcal{L}_1$ -penalized linear SVM
raising Hamming loss from 0.0090 to 0.0506.
The classes are evidently not linearly separable in the 22-dimensional MFCC

space

The $\mathcal{L}_1$ penalty additionally drives many weights to zero

The optimal C differs across labels

The degradation is largest for Family ($0.9931 \rightarrow 0.9393$) and smallest for Species ($0.9903 \rightarrow 0.9597$).

(biv)

Label	Best C	10-fold CV acc.	Resampled n
Family	10	0.9450	12372
Genus	10	0.9513	23232
Species	10	0.9534	24340
Test metric			Value
Exact match			0.8601
Hamming score			0.9301
Hamming loss			0.0699
Family / Genus / Species acc.			0.9148 / 0.9166 / 0.9588

SMOTE degrades every aggregate metric: exact match $0.9171 \rightarrow 0.8601$, Hamming loss $0.0506 \rightarrow 0.0699$.

Taking every class to the size of the largest inflates the training set from 5036 to 24340 samples for Species.

The synthetic points are interpolations between minority neighbors that need not lie in the true minority region, and for a linear classifier they pull the boundary away from the majority classes.

Since SMOTE targets minority recall, not overall accuracy.

Species is the most imbalanced and unaffected label ($0.9597 \rightarrow 0.9588$).

(bv)

Test metric	Value
Exact match	0.9893
Hamming score	0.9907
Hamming loss	0.0093
Family / Genus / Species acc.	0.9931 / 0.9898 / 0.9893

Chaining Family $\rightarrow$ Genus $\rightarrow$ Species (each classifier receiving the previous predictions as extra features)

Raising exact match slightly above the independent one-vs-all RBF SVM (0.9893 vs. 0.9866) while leaving Hamming score essentially unchanged (0.9907 vs. 0.9910).

The labels carry almost no independent information

The chain turns that redundancy into a constraint:

Predicted Genus that contradict the predicted Family becomes unlikely once the latter is an input feature.

With base accuracy already near 0.99, there is little inconsistency left to remove.

(bvi)

Confusion matrices:

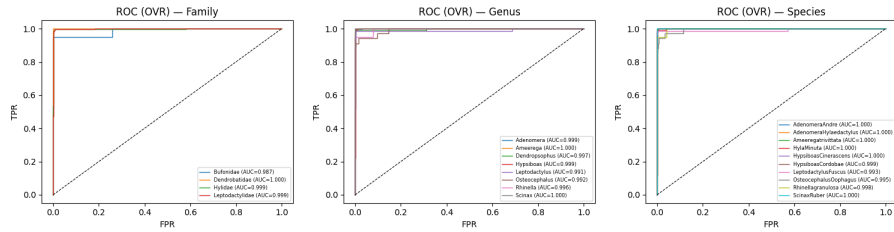
$$\text{Family} = \begin{pmatrix} 19 & 0 & 0 & 1 \\ 0 & 163 & 0 & 0 \\ 1 & 0 & 642 & 6 \\ 1 & 0 & 6 & 1320 \end{pmatrix}, \quad \text{Genus} = \begin{pmatrix} 1243 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 163 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 91 & 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 472 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 79 & 0 & 0 & 0 \\ 1 & 0 & 0 & 5 & 0 & 28 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 18 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 43 \end{pmatrix}$$

$$\text{Species} = \begin{pmatrix} 198 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1043 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 163 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 92 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 140 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 333 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 79 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 2 & 0 & 29 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 18 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 43 \end{pmatrix}$$

Aggregate scores:

Label	Accuracy	Macro P	Macro R	Macro F_1	Macro AUC	Micro AUC
Family	0.99	0.97	0.98	0.98	0.9961	0.9992
Genus	0.99	0.99	0.95	0.97	0.9969	0.9993
Species	0.99	0.98	0.97	0.97	0.9985	0.9996

ROC curves:



Explain: All curves hug the top-left corner.

micro weights each sample equally and is dominated by the large classes, macro weights each class equally and exposes the rare ones.

The lowest AUCs belong to the smallest classes

The same classes show the weakest recall (*Osteocephalus* 0.82, *Rhinella* 0.90)

Errors are structured, not random; confusion concentrates within the Hylidae family.

Precision stays ≥ 0.96 throughout while recall drops on rare classes, also the classifier is conservative about predicting a rare species.

Summary

Summary table is shown below:

	Exact match	Hamming score	Hamming loss
RBF normalized	0.986600	0.991000	0.009000
RBF raw	0.987500	0.992100	0.007900
L1 linear	0.917100	0.949400	0.050600
L1 + SMOTE	0.860100	0.930100	0.069900
Classifier Chain	0.989300	0.990700	0.009300